KENT PEARCE **MATH 1351 Spring 2004**

Homework Set HW2 due 2/2/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt) Problem 29, Section 2.1 page 60.
Evaluate the limit

$$\lim_{x\to 3} \frac{x^2+3x-10}{x-2}$$

2.(1 pt) Problem 5, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 1} \frac{x^2+x-3}{x+1}$$

3.(1 pt) Problem 8, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to \pi/4} \frac{1+\tan(x)}{\csc(x)+2}$$

4.(1 pt) Problem 12, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 2} \frac{x^2-4x+4}{x^2-x-2}$$

5.(1 pt) Problem 13, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 1} \frac{\frac{1}{x}-1}{x-1}$$

6.(1 pt) Problem 14, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 0} \frac{(x+1)^2-1}{x}$$

7.(1 pt) Problem 18, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 2} \frac{\sqrt{x+2}-2}{x-2}$$

8.(1 pt) Problem 20, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 0} \frac{\sqrt{x^2+4}-2}{x}$$

9.(1 pt) Problem 22, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 0^+} \frac{1-\cos(\sqrt{x})}{x}$$

10.(1 pt) Problem 24, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 0} \frac{\sin(4x)}{9x}$$

11.(1 pt) Problem 26, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 0} \frac{\cot(3x)}{\cot(x)}$$

12.(1 pt) Problem 28, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 0} \frac{\sin^2(x)}{2x}$$

13.(1 pt) Problem 30, Section 2.2 page 68.
Evaluate the limit

$$\lim_{x\to 0} \frac{x^2\cos(2x)}{1-\cos(x)}$$

14.(1 pt) Evaluate the limit

$$\lim_{t\to 8} \frac{\frac{1}{t}-\frac{1}{8}}{t-8}$$

15.(1 pt) Evaluate the limit

$$\lim_{a\to 1} \frac{a^3-a}{a^2-1}$$